

# Concentric Zero-Spheres of a class of sections of the commutative ring $\mathcal{R}$ of slice preserving functions over the compactified Octonions with the Riemann Hypothesis as a corollary

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*“...a ‘viewpoint’ by itself remains fragmentary. It reveals to us one of the aspects of a scenery or panorama, among a multiplicity of others which are equally valuable, equally ‘real’. It is when complementary viewpoints of a common reality are conjugated, that is, when our ‘eyes’ are multiplied, that the gaze is able to penetrate further ahead in the reckoning of things. ...it also happens, sometimes, that a sheaf of viewpoints converging to a unique and vast scenery gives rise to a novel thing; a thing which transcends each of the partial perspectives, in the same way that a living being transcends each of its limbs and organs. This new thing could be called a vision.”*

— A. Grothendieck, *Récoltes et Semailles*

## Abstract

Let  $\mathcal{R}$  be the commutative ring of slice-preserving slice-regular functions on the compactified octonions  $\mathbb{O}^* = S^8$ , under the regular  $(*)$  product. We prove the *Concentricity Theorem* for a section  $A \in \mathcal{R}$  satisfying four conditions: (C1) meromorphic continuation with a single simple pole carried to  $\infty = N$ ; (C2) an infinite Euler product; (C3) an infinite slice-regular Weierstrass factorization, whose spherical factors are the residue- $\mathbb{C}$  zeros; and (C4) infinitely many residue- $\mathbb{C}$  zeros. From the action groupoid  $\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*$ , whose non-real orbits are 6-spheres, the slice world  $\mathcal{S}_2$  of slice Riemann spheres with the band  $U(1)$ , and the section functor  $\Phi : \mathcal{H}_1 \rightarrow \mathcal{S}_2$ , we form the base  $\mathcal{B}$  of the exponential’s degenerate set  $\mathcal{B} \subset \mathcal{S}_2$  and the Grothendieck construction  $\mathcal{T}_A = \int_{\mathcal{B}} F$  of the band  $F = U(1)$  over  $\mathcal{B}$ , whose cocartesian transport is the winding lift. The theorem is that the residue- $\mathbb{C}$  zero 6-spheres of  $A$  lie in a single connected component of  $\mathcal{T}_A$ , arising as the degenerate fibre of its transport. The conclusion is computed at  $\pi_0$  by cocartesian diagram chasing, through the identity  $\pi_0(\mathcal{T}_A) \cong \text{colim}_{\mathcal{B}}(\pi_0 \circ F)$ ; the analytic slice-preserving inputs enter as cited statements. As downstream corollaries, the nontrivial-zero 6-spheres of the octonionic zeta function  $\zeta_{\mathbb{O}^*}$  are concentric about one real centre  $c$ ; hence every classical nontrivial zero has real part  $c$ ; and the functional equation forces  $c = \frac{1}{2}$ , giving the Riemann Hypothesis. Nothing in this document has yet been formalized in Lean.

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## Part I

# Classical Background

## 1 The Riemann zeta function

**Theorem 1.1** (Meromorphic continuation and functional equation; Riemann [13]). *The Dirichlet series  $\zeta(s) = \sum_{n \geq 1} n^{-s}$ , convergent for  $\operatorname{Re} s > 1$ , continues to a meromorphic function on  $\mathbb{C}$  with a single simple pole, at  $s = 1$ ; equivalently, a holomorphic map to the Riemann sphere*

$$\zeta : \mathbb{C}^* \longrightarrow \mathbb{C}^*, \quad \mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2, \quad \zeta(1) = \infty.$$

*The completed function  $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$  is entire and satisfies*

$$\xi(s) = \xi(1-s), \quad \xi(\bar{s}) = \overline{\xi(s)}.$$

*Consequently, if  $\rho$  is a nontrivial zero of  $\zeta$  then so are  $\bar{\rho}$ ,  $1-\rho$ , and  $1-\bar{\rho}$ . The trivial zeros of  $\zeta$  are at  $s = -2, -4, \dots$ ; the nontrivial zeros lie in the strip  $0 < \operatorname{Re} s < 1$ . The Riemann Hypothesis asserts that every nontrivial zero has  $\operatorname{Re} s = \frac{1}{2}$ .*

**Theorem 1.2** (Infinite Euler product; [15, Ch. 1]). *For  $\operatorname{Re} s > 1$ ,*

$$\zeta(s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1} = \exp\left(\sum_p \sum_{k \geq 1} \frac{1}{k} p^{-ks}\right),$$

*the product taken over the infinitely many primes; it converges absolutely and is zero-free on  $\operatorname{Re} s > 1$ .*

**Theorem 1.3** (Infinite Hadamard product; [15, Ch. 2]).  $\xi$  is entire of order 1 with infinitely many zeros, which are exactly the nontrivial zeros  $\{\rho\}$  of  $\zeta$ , and admits the Hadamard factorization

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho},$$

the product taken over the infinitely many nontrivial zeros.

**Corollary 1.4** (Infinitude of the nontrivial zeros).  $\zeta$  has infinitely many nontrivial zeros.

*Proof.* Immediate from Theorem 1.3: its zeros are infinitely many and are exactly the nontrivial zeros of  $\zeta$ .  $\square$

**Theorem 1.5** (Hardy [11]). Infinitely many nontrivial zeros of  $\zeta$  lie on the line  $\operatorname{Re} s = \frac{1}{2}$ .

## The compactified Riemann zeta on $\mathbb{C}^*$

For the slice construction,  $\zeta$  is taken in compactified form, with values on the Riemann sphere  $\mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2$  (the one-point compactification of  $\mathbb{C}$  [12]). The classical facts above are retained as stated.

**Definition 1.6** (Compactified classical zeta).  $\zeta_{\mathbb{C}^*} : \mathbb{C}^* \rightarrow \mathbb{C}^*$  is the meromorphic continuation of  $\zeta$  regarded as a map to the Riemann sphere:  $\zeta_{\mathbb{C}^*}(s) = \zeta(s)$  for  $s \in \mathbb{C} \setminus \{1\}$ ;  $\zeta_{\mathbb{C}^*}(1) = \infty$  (the simple pole, Theorem 1.1); and  $\zeta_{\mathbb{C}^*}(\infty) = 1$ , since  $\zeta(s) \rightarrow 1$  as  $\operatorname{Re}(s) \rightarrow +\infty$  (the  $p$ -test: only the  $n = 1$  term of  $\sum_n n^{-s}$  survives). It is holomorphic into  $\mathbb{C}^*$  on  $\mathbb{C}^* \setminus \{1\}$ , its sole pole being  $s = 1$ . The functional equation, the conjugate symmetry  $\zeta_{\mathbb{C}^*}(\bar{s}) = \overline{\zeta_{\mathbb{C}^*}(s)}$ , and Hardy's infinitude (Theorem 1.5) hold verbatim for  $\zeta_{\mathbb{C}^*}$ , as they concern values on  $\mathbb{C} \setminus \{1\}$  where  $\zeta_{\mathbb{C}^*} = \zeta$ .

**Lemma 1.7** (Compactification does not move the zeros).  $Z(\zeta_{\mathbb{C}^*}) = Z(\zeta)$ : the zeros of  $\zeta_{\mathbb{C}^*}$  in  $\mathbb{C}^*$  are exactly the zeros of  $\zeta$  in  $\mathbb{C}$ ; in particular the nontrivial zeros coincide.

*Proof.* On  $\mathbb{C} \setminus \{1\}$  the two functions agree, so they share all zeros there. The two adjoined values are not zeros:  $\zeta_{\mathbb{C}^*}(1) = \infty$  (a pole) and  $\zeta_{\mathbb{C}^*}(\infty) = 1$ ; and  $s = 1$ , being a pole of  $\zeta$ , was never a zero. Hence no zero is gained or lost.  $\square$

## Part II

# Slice-Preserving Theory, the Ring $\mathcal{R}$ , and the Equivalence

## 2 Octonions and slices

The octonions  $\mathbb{O}$  form an 8-dimensional real algebra with basis  $\{1, e_1, \dots, e_7\}$ , each  $e_i^2 = -1$ .

**Definition 2.1.** The octonions are: (i) a *division algebra*: every nonzero element has a multiplicative inverse; (ii) a *normed algebra*:  $|ab| = |a||b|$ ; (iii) *non-associative*:  $(ab)c \neq a(bc)$  in general; (iv) *alternative*:  $(aa)b = a(ab)$  and  $a(bb) = (ab)b$ .

An octonion  $s \in \mathbb{O}$  decomposes as  $s = \sigma + w$  with  $\sigma = \operatorname{Re}(s) \in \mathbb{R}$  and  $w = \operatorname{Im}(s) \in \operatorname{Im}(\mathbb{O}) \cong \mathbb{R}^7$ ; the unit imaginary octonions form  $S^6 \subset \operatorname{Im}(\mathbb{O})$ .

**Theorem 2.2** (Artin [14]). *In an alternative algebra, the subalgebra generated by any two elements is associative.*

**Corollary 2.3.** *Powers  $s^n$  are well-defined for any  $s \in \mathbb{O}$ .*

**Definition 2.4** (Complex slices and slice Riemann spheres). For any unit imaginary octonion  $v \in S^6$ , the *complex slice* is  $\mathbb{C}_v = \{a + bv : a, b \in \mathbb{R}\} \subset \mathbb{O}$ . Since  $v^2 = -1$ , this is a subalgebra isomorphic to  $\mathbb{C}$  via the  $\mathbb{R}$ -algebra isomorphism  $\varphi_v : \mathbb{C} \rightarrow \mathbb{C}_v, i \mapsto v$ . All slices share the real axis:  $\mathbb{C}_v \cap \mathbb{C}_w = \mathbb{R}$  for  $v \neq \pm w$ . Its one-point compactification is the *slice Riemann sphere*  $\mathbb{C}_v^* := \mathbb{C}_v \cup \{\infty\} = S_v^2$ , and  $\varphi_v$  extends to an isomorphism of Riemann spheres  $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$  with  $\varphi_v(\infty) = \infty$  (GPV [17], Prop. 4.1/Thm. 4.2). All slice spheres share the single point at infinity, as well as the real axis.

### 3 Slice-preserving functions

The class we use is the *slice-preserving* slice-regular functions. We first recall slice regularity [8, 6, 5, 9] — holomorphicity on each complex slice separately — as the ambient notion, then isolate the slice-preserving subclass, on which classical complex methods apply slicewise (by Artin’s theorem each slice  $\mathbb{C}_v$  is an associative subalgebra where classical complex analysis applies unchanged).

**Definition 3.1** (Slice regularity; [5, Def. 5.1.1]). Let  $\Omega \subseteq \mathbb{O}$  be open. A function  $f : \Omega \rightarrow \mathbb{O}$  is *slice regular* if for each  $v \in S^6$ ,  $\varphi_v^{-1} \circ f \circ \varphi_v$  satisfies the Cauchy–Riemann equations on  $\varphi_v^{-1}(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}$ . A domain  $\Omega$  is *axially symmetric* if  $\sigma + \gamma v \in \Omega$  implies  $\sigma + \gamma w \in \Omega$  for all  $w \in S^6$ .

**Theorem 3.2** (Representation Formula; [5, Thm. 5.1.7]). *Let  $f$  be slice regular on an axially symmetric slice domain  $\Omega$ . Fix  $v_0 \in S^6$  and write  $f(\sigma + \gamma v_0) = \alpha(\sigma, \gamma) + v_0 \cdot \beta(\sigma, \gamma)$  with  $\alpha, \beta : \mathbb{R}^2 \rightarrow \mathbb{R}$ . Then for every  $v \in S^6$ ,  $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v \cdot \beta(\sigma, \gamma)$ , where  $\alpha = \frac{1}{2}[f(\sigma + \gamma v_0) + f(\sigma - \gamma v_0)]$  and  $\beta = \frac{1}{2}v_0^{-1}[f(\sigma - \gamma v_0) - f(\sigma + \gamma v_0)]$ .*

**Theorem 3.3** (Identity Theorem; [5, Cor. 5.1.9]). *Let  $f$  be slice regular on an axially symmetric slice domain  $\Omega$ . If  $f$  vanishes on a subset of a single slice  $\Omega \cap \mathbb{C}_{v_0}$  with an accumulation point in  $\Omega \cap \mathbb{C}_{v_0}$ , then  $f \equiv 0$  on all of  $\Omega$ .*

**Theorem 3.4** (Extension Theorem; [5, Thm. 5.1.5]). *Let  $U \subseteq \mathbb{C}$  be a domain symmetric under conjugation and  $g : U \rightarrow \mathbb{C}$  holomorphic. There is a unique slice regular  $\text{ext}(g) : \Omega_U \rightarrow \mathbb{O}$  on  $\Omega_U = \{\sigma + \gamma v : \sigma + i\gamma \in U, v \in S^6\}$  with  $\text{ext}(g)|_{\mathbb{C}_{v_0}} = \varphi_{v_0} \circ g \circ \varphi_{v_0}^{-1}$  for every  $v_0$ .*

**Definition 3.5** (Slice-preserving functions; [1, Def. 2.7, Rem. 2.8]; [17, 7, 16]). A slice-regular function  $f$  on an axially symmetric domain  $\Omega$  is *slice preserving* if it maps every slice into itself:  $f(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}_v$  for all  $v \in S^6$ . For octonions ( $\#S^6 > 2$ ) the following are equivalent characterisations:

- (i)  $f$  is *intrinsic*:  $f(q^c) = f(q)^c$  for all  $q \in \Omega$ , where  $q^c$  is octonionic conjugation (on a slice,  $a + bv \mapsto a - bv$ );
- (ii) its representation-formula components are  $\mathbb{R}$ -valued: writing  $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v \beta(\sigma, \gamma)$  (Theorem 3.2), one has  $\alpha, \beta : \mathbb{R}^2 \rightarrow \mathbb{R}$  — equivalently the inducing stem function  $F = F_1 + \iota F_2$  has  $\mathbb{R}$ -valued components  $F_1, F_2$ ;
- (iii)  $f$  preserves two distinct slices; equivalently  $f(\Omega \cap \mathbb{R}) \subseteq \mathbb{R}$  when  $\Omega$  is a slice domain.

Slice-preserving functions are the more restrictive subclass of slice-regular functions on which classical complex methods apply slicewise. They are precisely the sections of  $\mathcal{R}$  (Definition 5.1), and on them the regular product is the ordinary pointwise product, so the product is commutative (Theorem 5.2; Ghiloni–Perotti–Stoppato [7, Rem. 1.8]).

*Remark 3.6* (Compactified extension). Definition 2.4 through Definition 3.5, and Theorems 3.2–3.4, are stated in the literature on uncompactified domains  $\Omega \subseteq \mathbb{O}$  (and  $U \subseteq \mathbb{C}$ ). We use them on the compactified  $\mathbb{O}^* = S^8$  and on the slice Riemann spheres  $\mathbb{C}_v^* = S_v^2$  (GPV’s slice Riemann manifolds [17]). The passage  $\mathbb{O} \rightsquigarrow \mathbb{O}^*$ ,  $\mathbb{C}_v \rightsquigarrow \mathbb{C}_v^*$  adjoins only the shared point  $\infty = N$ ; it is recorded as a marked extension node (in the Lean development: cite the uncompactified statement and extend through the one-point compactification `OnePoint`).

## 4 The octonionic zeta function $\zeta_{\mathbb{O}^*}$

**Definition 4.1** (Octonionic zeta). Here  $N$  denotes the *north pole* of  $S^8$ : the single point at infinity  $\infty$  adjoined in the one-point compactification  $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} \cong S^8$  (inverse stereographic projection; GPV [17]). It is one point with two names, so  $\infty = N$ ; it is the only point at infinity, shared by  $\mathbb{O}^*$  and by every slice Riemann sphere  $\mathbb{C}_v^* = S_v^2$ .

The *octonionic zeta function*  $\zeta_{\mathbb{O}^*} : \mathbb{O}^* \rightarrow \mathbb{O}^*$  is defined slicewise from the compactified classical zeta (Definition 1.6) through the slice identifications  $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^* = S_v^2$  (the  $\mathbb{R}$ -algebra isomorphism  $\mathbb{C} \rightarrow \mathbb{C}_v$  of Definition 2.4, extended to the Riemann spheres by  $\varphi_v(\infty) = N$ ; GPV [17], Prop. 4.1/Thm. 4.2):

- (i) for  $s \in \mathbb{O} \setminus \mathbb{R}$ , with  $w = \text{Im}(s)$ ,  $v = w/|w| \in S^6$ ,  $\gamma = |w| > 0$ ,  $\sigma = \text{Re}(s)$ , so  $s = \sigma + \gamma v$ :  
 $\zeta_{\mathbb{O}^*}(s) := \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$ ;
- (ii) for  $s \in \mathbb{R} \setminus \{1\}$ :  $\zeta_{\mathbb{O}^*}(s) := \zeta_{\mathbb{C}^*}(s) = \zeta(s) \in \mathbb{R}$ ;
- (iii) for  $s = 1$  (the simple pole, Definition 1.6):  $\zeta_{\mathbb{O}^*}(1) := \infty = N$ ;
- (iv) for  $s = \infty = N$ :  $\zeta_{\mathbb{O}^*}(\infty) := \zeta_{\mathbb{C}^*}(\infty) = 1$ .

**Proposition 4.2** (Well-definedness).  $\zeta_{\mathbb{O}^*} : \mathbb{O}^* \rightarrow \mathbb{O}^*$  is well-defined.

*Proof.* For  $s \in \mathbb{O} \setminus \mathbb{R}$  the decomposition  $s = \sigma + \gamma v$  with  $\gamma > 0$ ,  $v \in S^6$  is forced ( $\gamma = |\text{Im } s|$ ,  $v = \text{Im } s / |\text{Im } s|$ ), so the only ambiguity is the identification of the slice  $\mathbb{C}_v = \mathbb{C}_{-v}$  with  $\mathbb{C}^*$ . That slice carries two such identifications,  $\varphi_v (i \mapsto v)$  and  $\varphi_{-v} (i \mapsto -v)$ , under which  $s$  reads as  $\sigma + i\gamma$  resp.  $\sigma - i\gamma$ ; we must check  $\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)) = \varphi_{-v}(\zeta_{\mathbb{C}^*}(\sigma - i\gamma))$ . By conjugate symmetry  $\zeta_{\mathbb{C}^*}(\sigma - i\gamma) = \overline{\zeta_{\mathbb{C}^*}(\sigma + i\gamma)}$  ([15, Ch. 2]; Definition 1.6), writing  $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = a + bi$  gives  $\varphi_{-v}(a - bi) = a + (-b)(-v) = a + bv = \varphi_v(a + bi)$ : the two sign reversals — one from conjugating the input, one from the reversed identification — cancel. The remaining cases carry no ambiguity: for real  $s \neq 1$ ,  $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$  for every  $v$ ; at  $s = 1$  the value is the single point  $\infty = N$ ; and at  $s = \infty = N$  the value is  $\zeta_{\mathbb{C}^*}(\infty) = 1 \in \mathbb{R}$ , again the same for every  $v$  since each  $\varphi_v$  fixes  $\mathbb{R}$  pointwise.  $\square$

*Remark 4.3* (Domain, codomain, and the point at infinity). Both domain and codomain are  $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} \cong S^8$ , the one-point (Alexandroff) compactification of  $\mathbb{O} \cong \mathbb{R}^8$  [12]. The map  $\zeta_{\mathbb{O}^*}$  is slice regular on  $\mathbb{O}^* \setminus \{1, \infty\}$  (Theorem 7.2); the simple pole at  $s = 1$  is carried to the point at infinity  $\infty = N$ , which is the *image* of the pole, not a zero. We write  $\mathbb{O}^*$  throughout; it is the one-point compactification, and is deliberately **not** a projective space  $\mathbb{O}P^1$  (there is no quotient by scalars).

## 5 $\mathcal{R}$ is a commutative ring

We work on  $\mathbb{O}^*$ ; write  $\Omega_v := \mathbb{O}^* \cap \mathbb{C}_v = \mathbb{C}_v^* = S_v^2$  for the slice Riemann sphere (Definition 2.4).

**Definition 5.1** (The ring of slice-preserving functions under the regular product).  $\mathcal{R} := \mathcal{O}_{\mathbb{O}^*}$  is the set of all slice-regular functions  $f : \mathbb{O}^* \rightarrow \mathbb{O}^*$  that are slice preserving (Definition 3.5),  $f(\Omega_v) \subseteq \mathbb{C}_v^*$  for all  $v \in S^6$ , equipped with pointwise addition and the regular  $(*)$  product.

Commutativity of  $\mathcal{R}$  rests on a single slice-preserving fact from the literature: on each slice the regular product is ordinary multiplication.

**Theorem 5.2** (Regular product on slice-preserving functions; [16, Rem. 2.11]). *For slice-preserving  $f, g$  (with  $f(\Omega_v) \subseteq \mathbb{C}_v$ ), the regular product restricts on each slice to the pointwise product,  $(f * g)|_{\Omega_v}(w) = f(w) \cdot g(w) \in \mathbb{C}_v$ , and  $f * g = g * f$ ,  $(f * g) * h = f * (g * h)$  (via Artin, Theorem 2.2).*

**Proposition 5.3.**  $(\mathcal{R}, +, *)$  is a commutative ring with identity  $1 \in \mathcal{R}$ .

*Proof.* Closure under  $+$  is pointwise. For  $*$ : fixing a slice  $\mathbb{C}_K$ , the product reduces to pointwise multiplication on  $\mathbb{C}_K$  by Theorem 5.2, which is commutative because  $\mathbb{C}_K$  is a commutative associative subalgebra; associativity and distributivity are inherited slicewise, and a slice-regular function is determined by its restriction to any one slice (Theorem 3.3). The constant 1 is the identity.  $\square$

**Proposition 5.4** ( $\mathcal{R}$  is an integral domain). *If  $f, g \in \mathcal{R}$  and  $f * g = 0$ , then  $f = 0$  or  $g = 0$ .*

*Proof.*  $(f * g)|_{\Omega_K}(w) = f(w) \cdot g(w)$  in  $\mathbb{C}_K \cong \mathbb{C}$  (Theorem 5.2). The restrictions  $f|_{\Omega_K}, g|_{\Omega_K}$  are holomorphic on the connected slice; by the classical identity principle the pointwise product vanishes identically iff one factor does. By Theorem 3.3 the corresponding global function vanishes.  $\square$

## 6 The slice-preserving analytic toolkit on the compactified octonions

We collect the slice-preserving analytic facts used in Part 3 as cited statements: the exponential, its logarithm manifold, its degenerate fibre, the slice/stem square and  $I$ -independent lift, and the winding lift. Each is established in the literature on the uncompactified  $\mathbb{O}$  and used here on  $\mathbb{O}^* = S^8$  and the slice Riemann spheres  $\mathbb{C}_v^* = S_v^2$  (the passage is recorded in Remark 3.6; the compactified slice-manifold structure itself is GPV's [17]).

**Theorem 6.1** (The slice exponential; [21, §3]; [17, Prop. 5.1]; [1, Rem. 2.23]). *The exponential  $\exp : \mathbb{O} \rightarrow \mathbb{O} \setminus \{0\}$ ,  $\exp(q) = \sum_{k \geq 0} q^k / k!$  (powers well defined by Corollary 2.3), is a slice-regular, slice-preserving entire function (Definition 3.5), given on each slice by*

$$\exp(x + Iy) = e^x(\cos y + I \sin y), \quad x, y \in \mathbb{R}, \quad I \in S^6.$$

*Consequently  $|\exp q| = e^{\operatorname{Re} q}$  depends only on the real part, and  $\exp$  is nowhere zero.*

**Theorem 6.2** (The logarithm manifold  $E_{\mathbb{O}}^+$ ; [17, Prop. 5.1, Rem. 5.2, Def. 5.3, Prop. 5.4, Def. 5.5]; [21, §3]). *Let  $T : \mathbb{O} \rightarrow \mathbb{O} \times \operatorname{Im} \mathbb{O}$ ,  $T(x + Iy) = (\sinh x \cos y + I \sinh x \sin y, Iy)$ , and  $\mathbb{O}^+ = \{q : \operatorname{Re} q > 0\}$ . The logarithm manifold is  $E_{\mathbb{O}}^+ := T(\mathbb{O}^+)$ , the semi-helicoidal hypercomplex 8-manifold; equivalently it is the image of the exponential graph, parametrised by the  $E_{\mathbb{O}}^+$ -exponential*

$$E : \mathbb{O} \longrightarrow E_{\mathbb{O}}^+ \subset \mathbb{O} \times \operatorname{Im} \mathbb{O}, \quad E(x + Iy) = (\exp(x + Iy), Iy) = (e^x \cos y + Ie^x \sin y, Iy),$$

an immersion and a diffeomorphism, with inverse the  $E_{\mathbb{O}}^+$ -logarithm  $L : E_{\mathbb{O}}^+ \rightarrow \mathbb{O}$ ,  $L(q, p) = \log |q| + p$  (here  $p = \text{Arg}(q)$ ). Writing  $\pi$  for the projection to the first factor,  $\pi \circ E = \exp$  (Theorem 6.1); but  $\pi : E_{\mathbb{O}}^+ \rightarrow \mathbb{O} \setminus \{0\}$  is not a covering and not an open map, precisely because  $\exp$  is not open. Thus  $E_{\mathbb{O}}^+$  is an adapted blow-up of  $\mathbb{O}$  along the real axis: it unwraps the multivalued logarithm into the single-valued  $L$ , and a branch of the hypercomplex logarithm is  $\log_{\mathbb{O}} = L \circ (\pi|_{\Omega})^{-1}$  on any path-connected  $\Omega \subset E_{\mathbb{O}}^+$  on which  $\pi|_{\Omega}$  is injective.

**Lemma 6.3** (The degenerate set of the exponential; its fibre over a real value). *Sourced.* With  $\pi$  the first-factor projection,  $\pi \circ E = \exp$  ([17, Rem. 5.2(a)] — the commuting triangle); and, unlike the complex setting,  $\pi : E_{\mathbb{O}}^+ \rightarrow \mathbb{O}$  is not a covering and not an open map, because  $\exp$  is not open: it has a non-empty degenerate set consisting of spheres ([17, Rem. 5.2(b)], verbatim). The direction  $I(q)$  admits no continuous extension to any point of  $\mathbb{R}$  ([21, Rem. 2.1]).

*Proved from the slice form. For  $r > 0$ ,*

$$\exp^{-1}(-r) = \{ \log r + I(2k+1)\pi : I \in S^6, k \in \mathbb{Z} \} :$$

by Theorem 6.1,  $\exp(x + Iy) = e^x \cos y + Ie^x \sin y = -r$  forces  $e^x \sin y = 0$ , hence  $y \in \pi\mathbb{Z}$ ; negativity of the value forces  $y = (2k+1)\pi$  and  $x = \log r$ ; the direction  $I$  is unconstrained. The fibre is thus indexed by the single real level  $\log r = \log |-r|$  and, within it, by the odd winding indices  $2k+1$  carried as band data (the winding lift, Theorem 6.6; [21, Cor. 5.21]). The fibre formula itself appears, stated without proof as Preface motivation, in [17] (p. 972); the slice-form derivation above is the load-bearing statement.

*Remark 6.4* (“Concentric” is translation vocabulary). Geometrically the fibre of Lemma 6.3 is a family of 6-spheres of radii  $\pi, 3\pi, 5\pi, \dots$  sharing the centre  $\log r$  — concentric. The construction of Part 3 never uses this metric fact: the index relations enter only through the commuting triangle  $\pi \circ E = \exp$ , the level, and the winding. The word belongs to the downstream translation (Corollary 10.9), where one component of the static base is read as one real centre.

**Definition 6.5** (Slice restriction and the  $I$ -independent lift; [16, Rem. 2.11]; [4, §3.3, §3.7]; [7, Def. 2.1, Prop. 2.2]). A slice-preserving section  $A$  carries each slice Riemann sphere into itself,

$$A(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^* = S_v^2 \quad (v \in S^6),$$

so its restriction to a slice is a holomorphic self-map of that slice sphere,

$$A_v := \varphi_v^{-1} \circ A \circ \varphi_v : \mathbb{C}^* \longrightarrow \mathbb{C}^*,$$

through the identification  $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$  of Definition 2.4. The  $I$ -independent lift is the fact that this holomorphic stem is the *same* for every  $v$ : there is one holomorphic  $A_{\circ} : \mathbb{C}^* \rightarrow \mathbb{C}^*$  with

$$A|_{\mathbb{C}_v^*} = \varphi_v \circ A_{\circ} \circ \varphi_v^{-1} \quad \text{for all } v \in S^6$$

(Wang [16, Rem. 2.11]; Bisi–Winkelmann [4, §3.7]). Equivalently,  $A$  is *intrinsic*,  $A(q^c) = A(q)^c$ , and is induced ( $A = \mathcal{I}(F)$ ) by a single stem function  $F = F_1 + \iota F_2 : D \rightarrow \mathbb{O}_{\mathbb{C}}$  on the symmetric  $D \subset \mathbb{C}$  whose components  $F_1, F_2$  are  $\mathbb{R}$ -valued (the stem-function formalism, Ghiloni–Perotti [7, Def. 2.1, Prop. 2.2]):

$$A(x + Iy) = F_1(x + iy) + I F_2(x + iy), \quad F_1, F_2 \text{ } \mathbb{R}\text{-valued.}$$

It is this single  $I$ -independent slice stem that makes the section functor  $\Phi$   $G_2$ -equivariant, hence blind to the  $G_2$ -orbit direction (Definition 9.4, Theorem 9.7).

**Theorem 6.6** (The winding lift; [21, Def. 3.4, Def. 4.1, Prop. 4.2, Def. 5.11]). *Let  $\gamma : [a, b] \rightarrow \mathbb{O} \setminus \{0\}$  be a path (on each slice  $\mathbb{C}_v$  this is the classical complex continuation). A continuation of the hypercomplex logarithm along  $\gamma$  is a path  $\tilde{\gamma} : [a, b] \rightarrow \mathbb{O}$  with  $\exp \circ \tilde{\gamma} = \gamma$ , and a lift of  $\gamma$  to the logarithm manifold  $E_{\mathbb{O}}^+$  (Theorem 6.2) is a path  $\Gamma : [a, b] \rightarrow E_{\mathbb{O}}^+$  with  $\text{pr}_1 \circ \Gamma = \gamma$ :*

$$\begin{array}{ccc} & \nearrow \tilde{\gamma} & \mathbb{O} \\ [a, b] & \xrightarrow{\gamma} & \mathbb{O} \setminus \{0\} \\ & \uparrow \exp & \end{array} \qquad \begin{array}{ccc} & \nearrow \Gamma & E_{\mathbb{O}}^+ \\ [a, b] & \xrightarrow{\gamma} & \mathbb{O} \setminus \{0\} \\ & \downarrow \text{pr}_1 & \end{array}$$

The two data are equivalent: a lift  $\Gamma$  exists if and only if a continuation  $\tilde{\gamma}$  exists, and they correspond by

$$\tilde{\gamma} = L \circ \Gamma, \quad \Gamma = (\exp \circ \tilde{\gamma}, \text{Im } \tilde{\gamma}),$$

where  $L$  is the  $E_{\mathbb{O}}^+$ -logarithm of Theorem 6.2 ([21, Prop. 4.2]). For a loop  $\gamma : S^1 \rightarrow \mathbb{O} \setminus \{0\}$  the lift is taken on the universal cover  $\exp : i\mathbb{R} \rightarrow S^1$ , as a continuous  $\Gamma : i\mathbb{R} \rightarrow E_{\mathbb{O}}^+$  with  $\text{pr}_1 \circ \Gamma = \gamma \circ \exp$  ([21, Def. 5.11]):

$$\begin{array}{ccc} i\mathbb{R} & \xrightarrow{\Gamma} & E_{\mathbb{O}}^+ \\ \exp \downarrow & & \downarrow \text{pr}_1 \\ S^1 & \xrightarrow{\gamma} & \mathbb{O} \setminus \{0\} \end{array}$$

**Proposition 6.7** (Tameness, existence, and the winding number; [21, Def. 4.20, Cor. 5.21, Cor. 5.22]).

The lift of Theorem 6.6 is governed by the companion structure of  $\gamma$ .

- (i) *Uniqueness (tameness).*  $\gamma$  is tame when it admits a unique companion imaginary-unit field; its lift is then unique ([21, Def. 4.20]).
- (ii) *Existence and closure.* A lift of a loop  $\gamma$  exists if and only if the signature satisfies  $\sigma(\gamma|_{[\xi_l, \xi_{l+1}]}) \in \{0, -1\}$  on each obstruction interval; and if it exists, the lift is itself a loop ([21, Cor. 5.13]; pinned against the arXiv text — the earlier citation to Cor. 5.22 is superseded).
- (iii) *Winding number.* For an untwisted tame loop with even circular signature  $\sigma^c$ , the winding number is  $\omega = |\sigma^c|/2$  ([21, Cor. 5.21]).

## 7 Slice preservation and symmetries

**Theorem 7.1** (Slice preservation).  $\zeta_{\mathbb{O}^*}$  is slice preserving (Definition 3.5):  $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$  for every  $v \in S^6$ .

*Proof.* Fix  $v \in S^6$  and  $s \in \mathbb{C}_v^*$ . If  $s \in \mathbb{C}_v \setminus \mathbb{R}$  then  $s = \sigma \pm \gamma v$  and  $\zeta_{\mathbb{O}^*}(s) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma \pm i\gamma)) \in \varphi_v(\mathbb{C}^*) = \mathbb{C}_v^*$  (Definition 4.1(i)). If  $s \in \mathbb{R} \setminus \{1\}$  then  $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$ ; while  $\zeta_{\mathbb{O}^*}(1) = \infty \in \mathbb{C}_v^*$  and  $\zeta_{\mathbb{O}^*}(\infty) = 1 \in \mathbb{C}_v^*$ . In every case the value remains in  $\mathbb{C}_v^*$ .  $\square$

**Theorem 7.2** ( $\zeta_{\mathbb{O}^*}$  is a section of  $\mathcal{R}$ ).  $\zeta_{\mathbb{O}^*} \in \mathcal{R}$ .

*Proof.* On each slice the restriction is the compactified classical zeta read through the isomorphism  $\varphi_v$ , namely  $\zeta_{\mathbb{O}^*}|_{\mathbb{C}_v^*} = \varphi_v \circ \zeta_{\mathbb{C}^*} \circ \varphi_v^{-1}$ , holomorphic into  $\mathbb{C}_v^*$  away from the single pole. Equivalently  $\zeta_{\mathbb{O}^*} = \text{ext}(\zeta)$  is the slice-regular extension of the holomorphic, conjugation-symmetric  $\zeta$  on  $\mathbb{C} \setminus \{1\}$  (Theorem 3.4, [5, Thm. 5.1.5]); hence  $\zeta_{\mathbb{O}^*}$  is slice regular on  $\mathbb{O}^* \setminus \{1, \infty\}$  (slice regularity is slice-wise

holomorphy, Definition 3.1). By Theorem 7.1 it is slice preserving,  $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$ . A slice-regular, slice-preserving function  $\mathbb{O}^* \rightarrow \mathbb{O}^*$  is by definition a section of  $\mathcal{R} = \mathcal{O}_{\mathbb{O}^*}$  (Definition 5.1); the regular product on such sections restricts on each slice to the ordinary product (Theorem 5.2, Wang [16, Rem. 2.11]). Finally  $\zeta_{\mathbb{O}^*}$  is not a unit, since it has zeros (Theorem 8.2).  $\square$

**Definition 7.3** (The automorphism group  $G_2$ ).  $G_2 = \text{Aut}(\mathbb{O})$  is the 14-dimensional compact, connected, simply connected Lie group of  $\mathbb{R}$ -algebra automorphisms of  $\mathbb{O}$ . Each  $g \in G_2$  fixes 1, hence the real axis pointwise, and acts on  $\text{Im } \mathbb{O} \cong \mathbb{R}^7$  as an element of  $SO(7)$ ; thus  $g \cdot (\sigma + \gamma v) = \sigma + \gamma g(v)$ .

**Theorem 7.4** ( $G_2$ -equivariance).  $\zeta_{\mathbb{O}^*}(g \cdot s) = g \cdot \zeta_{\mathbb{O}^*}(s)$  for all  $g \in G_2$ ,  $s \in \mathbb{O}$ .

*Proof.* For  $s \in \mathbb{R}$ ,  $g \cdot s = s$  and  $\zeta_{\mathbb{O}^*}(s) \in \mathbb{R}$  is fixed, so both sides equal  $\zeta_{\mathbb{O}^*}(s)$ . For a non-real  $s = \sigma + \gamma v$  ( $v \in S^6$ ,  $\gamma > 0$ ),  $g$  carries the slice  $\mathbb{C}_v$  to the slice  $\mathbb{C}_{g(v)}$  by the isometric relabelling  $\varphi_{g(v)} = g \circ \varphi_v$  (Definition 7.3). Since  $\zeta_{\mathbb{O}^*}$  is defined slicewise by the same  $\zeta_{\mathbb{C}^*}$  read through  $\varphi_v$  (Definition 4.1(i)),

$$\zeta_{\mathbb{O}^*}(g \cdot s) = \varphi_{g(v)}(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)) = g(\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))) = g \cdot \zeta_{\mathbb{O}^*}(s).$$

$\square$

**Theorem 7.5** ([3]).  $G_2$  acts transitively on  $S^6$  with stabilizer  $SU(3)$ :  $SU(3) \hookrightarrow G_2 \twoheadrightarrow S^6$ .

*Remark 7.6* (The  $G_2$ -action extends to  $\mathbb{O}^* = S^8$ ). The action above is stated on  $\mathbb{O}$ , but  $\mathcal{H}_1$  (Part 3) runs on the compactified  $\mathbb{O}^* = S^8$ . Each  $g \in G_2$  fixes the real axis pointwise (an automorphism fixes 1, hence all of  $\mathbb{R}$ ) and acts on  $\text{Im } \mathbb{O} = \mathbb{R}^7$  as an element of  $SO(7)$ ; in particular  $g$  is a linear isometry of  $\mathbb{R}^8 = \mathbb{O}$ , hence proper, and therefore extends uniquely to a homeomorphism of the one-point compactification  $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$  fixing the point at infinity  $\infty = N$  (the one-point compactification is functorial on proper maps; in Lean, `OnePoint`). Thus  $G_2 \curvearrowright \mathbb{O}^*$  fixes the compactified real axis  $\mathbb{R} \cup \{\infty\}$  — a great circle through  $N$  — pointwise, and acts on each direction-sphere  $S^6$  exactly as in Theorem 7.5. This is the action on which  $\mathcal{H}_1$  is built, with  $\mathcal{R}$  its ring of invariants over  $\mathbb{O}^*$ .

## 8 Zero geometry and the equivalence

**Theorem 8.1** (Zero Equivalence Theorem). Let  $\rho = \sigma + i\gamma \in \mathbb{C}^*$ . For every  $v \in S^6$ ,

$$\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = 0 \iff \zeta_{\mathbb{C}^*}(\rho) = 0.$$

*Equivalently:* a non-real  $s \in \mathbb{O}^*$  is a zero of  $\zeta_{\mathbb{O}^*}$  if and only if its slice coordinate  $\varphi_v^{-1}(s)$  is a zero of  $\zeta_{\mathbb{C}^*}$ ; by Lemma 1.7 these are exactly the nontrivial zeros of the classical  $\zeta$ .

*Proof.* By Definition 4.1(i),  $\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$ . The slice identification  $\varphi_v : \mathbb{C}^* \rightarrow \mathbb{C}_v^* = S_v^2$  is an isomorphism of slice Riemann spheres (the  $\mathbb{R}$ -algebra isomorphism  $\mathbb{C} \rightarrow \mathbb{C}_v$  of Definition 2.4, extended by  $\varphi_v(\infty) = N$ ; GPV [17]). An isomorphism is injective and sends 0 to 0, so  $\varphi_v(z) = 0 \iff z = 0$ ; with  $z = \zeta_{\mathbb{C}^*}(\rho)$  this gives both implications. (Slice preservation, Theorem 7.1, is what places the value in  $\mathbb{C}_v^*$  so that  $\varphi_v^{-1}$  applies on the non-real side.) The final clause is Lemma 1.7.  $\square$

**Theorem 8.2** (Nontrivial zeros are infinitely many disjoint 6-spheres). For a nontrivial zero  $\rho = \sigma + i\gamma$  of  $\zeta_{\mathbb{C}^*}$  (so  $\gamma > 0$ ) set

$$S_\rho = \{\sigma + \gamma v : v \in S^6\} = \sigma + \gamma S^6 \subset \mathbb{O}.$$

*Then:*

- (i)  $S_\rho$  is a 6-sphere — the round sphere of radius  $\gamma$  centred at the real point  $\sigma$  — equivalently the  $G_2$ -orbit of any one of its points  $\sigma + \gamma v_0$  (Theorem 7.4;  $G_2$  acts transitively on  $S^6$  with stabiliser  $\mathrm{SU}(3)$ , Theorem 7.5, Baez [3]).
- (ii) every point of  $S_\rho$  is a zero of  $\zeta_{\mathbb{O}^*}$ , and conjugate zeros give the same sphere,  $S_\rho = S_{\bar{\rho}}$ ;
- (iii) the nontrivial-zero set is the disjoint union  $Z_{\mathbb{O}^*}^{\mathrm{nt}} = \bigsqcup_{[\rho]} S_\rho$  over conjugate pairs  $[\rho] = \{\rho, \bar{\rho}\}$ , with distinct pairs giving disjoint spheres;
- (iv) there are infinitely many such spheres (Corollary 1.4; also Hardy [11], Theorem 1.5).

*Proof.* (i) The map  $v \mapsto \sigma + \gamma v$  is the affine map  $x \mapsto \sigma + \gamma x$  of  $\mathrm{Im} \mathbb{O} \cong \mathbb{R}^7$  restricted to  $S^6$ ; with  $\gamma > 0$  its image is the round 6-sphere of radius  $\gamma$  centred at  $\sigma$ . By  $G_2$ -equivariance (Theorem 7.4) and transitivity of  $G_2$  on  $S^6$  (Theorem 7.5),  $G_2 \cdot (\sigma + \gamma v_0) = \{\sigma + \gamma g(v_0) : g \in G_2\} = S_\rho$ . So  $S_\rho$  is  $S^6$  — scaled by  $\gamma$ , translated to  $\sigma$  — realized as a single  $G_2$ -orbit; no embedding or diffeomorphism is invoked.

(ii) For any  $v \in S^6$ ,  $\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\rho)) = \varphi_v(0) = 0$  by the Zero Equivalence Theorem 8.1. And  $S_{\bar{\rho}} = \{\sigma + \gamma(-v) : v \in S^6\} = S_\rho$ , since  $v \mapsto -v$  is a bijection of  $S^6$ .

(iii) For “ $\subseteq$ ”: a non-real zero  $s = \sigma + \gamma v$  of  $\zeta_{\mathbb{O}^*}$  forces  $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = 0$  by Theorem 8.1, so  $s \in S_\rho$  with  $\rho = \sigma + i\gamma$  nontrivial (Lemma 1.7); “ $\supseteq$ ” is (ii). Disjointness: if  $\sigma + \gamma v = \sigma' + \gamma' w$ , uniqueness of the real/imaginary parts gives  $\sigma = \sigma'$  and  $\gamma v = \gamma' w$ , whence  $\gamma = \gamma'$  (taking norms,  $|v| = |w| = 1$ ) and  $\{\rho', \bar{\rho}'\} = \{\rho, \bar{\rho}\}$ ; so two spheres coincide or are disjoint.

(iv) There are infinitely many spheres because  $\zeta$  has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5).  $\square$

**Theorem 8.3** (The equivalence: concentricity  $\Leftrightarrow$  RH). *The following are equivalent:*

- (a) the Riemann Hypothesis;
- (b) the nontrivial-zero 6-spheres  $\{S_\rho\}$  of  $\zeta_{\mathbb{O}^*}$  (Theorem 8.2) are concentric — they share a single common centre, necessarily a point of the real axis.

When they hold, the common centre is  $\frac{1}{2}$ .

*Proof.* Each  $S_\rho$  is centred at the real point  $\sigma_\rho$ , where  $\rho = \sigma_\rho + i\gamma_\rho$  (Theorem 8.2(i)).

(a)  $\Rightarrow$  (b): under RH every  $\sigma_\rho = \frac{1}{2}$ , so all spheres share the centre  $\frac{1}{2}$ .

(b)  $\Rightarrow$  (a): suppose the spheres share a common centre  $c \in \mathbb{R}$ . By the functional equation (Theorem 1.1), if  $\rho$  is a nontrivial zero then so is  $1 - \bar{\rho} = (1 - \sigma_\rho) + i\gamma_\rho$ , whose sphere  $S_{1-\bar{\rho}}$  is centred at  $1 - \sigma_\rho$ . Concentricity forces  $\sigma_\rho = c = 1 - \sigma_\rho$ , hence  $\sigma_\rho = \frac{1}{2}$ ; as  $\rho$  was arbitrary this is RH, and the common centre is  $\frac{1}{2}$ .  $\square$

*Remark 8.4* ( $\frac{1}{2}$  is downstream of concentricity). The value  $\frac{1}{2}$  enters *only* in (b)  $\Rightarrow$  (a), and only through the functional equation pinning the shared hyperplane once concentricity is known. Part 3 establishes the one-component collapse — that the residue- $\mathbb{C}$  zero spheres of  $A$  all lie in one connected component of the Grothendieck construction  $\mathcal{T}_A$  (Definition 9.9), concentricity then following by the translation of Corollary 10.9 — with no reference to  $\frac{1}{2}$ , to coordinates, or to RH; the value is then supplied here, downstream, by the classical functional equation alone.

## Part III

# The Categorical Construction and the Concentricity Theorem

We build the translation groupoid  $\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*$  (whose orbits of non-real points are the residue- $\mathbb{C}$  zero 6-spheres), the slice world  $\mathcal{S}_2$  (the codomain  $\mathbb{O}^*$  by its slice decomposition, carrying the band  $U(1)$  and the direction  $G_2$ ), the section functor  $\Phi : \mathcal{H}_1 \rightarrow \mathcal{S}_2$ , the base  $\mathcal{B}$  of the exponential's degenerate set, and the Grothendieck construction  $\mathcal{T}_A = \int_{\mathcal{B}} F$  of the band  $F = U(1)$  over  $\mathcal{B}$ . Hypotheses C1–C4 assemble the transport of  $\mathcal{T}_A$ ; the residue- $\mathbb{C}$  zero 6-spheres of  $A$  arrive as the degenerate fibre of that transport — an output, never an input — and the Concentricity Theorem is that they lie in a single connected component of  $\mathcal{T}_A$ , computed at  $\pi_0$ . The categorical objects are anchored to the literature (Quillen [24], Thomason [23], Riehl [25], Goerss–Jardine [20], Vistoli [26], Stacks [18]) and to Mathlib terms; the slice-preserving analytic facts of Part 2 enter as cited inputs.

## 9 Two worlds and the section between them

### The compactified octonions and the point at infinity

**Definition 9.1** (The compactified octonions  $\mathbb{O}^* = S^8$ ). Let  $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$  be the Alexandroff one-point compactification of  $\mathbb{O} = \mathbb{R}^8$ . Concretely (GPV [17], Prop. 4.1 and Thm. 4.2, for  $m = \dim_{\mathbb{R}} K \in \{4, 8\}$ ): the inverse stereographic projections from the north and south poles,

$$f(x + Iy) = \left( \frac{2(x + Iy)}{1 + x^2 + y^2}, \frac{-1 + x^2 + y^2}{1 + x^2 + y^2} \right), \quad h(x + Iy) = \left( \frac{2(x - Iy)}{1 + x^2 + y^2}, \frac{1 - x^2 - y^2}{1 + x^2 + y^2} \right),$$

give a conformal atlas  $\{(K, f), (K, h)\}$  endowing  $S^m$  with the structure of a slice (quaternionic/octonionic) Riemann manifold, with transition map  $h^{-1} \circ f(q) = \bar{q}/q^2 = 1/q$ , itself slice regular and slice preserving. The adjoined point is the north pole  $N$ ; this is the point at infinity. In Lean the underlying compactification is `Mathlib.Topology.Compactification.OnePoint`,  $\mathbb{O}^* = \text{OnePoint } \mathbb{O}$  with  $\infty = \text{OnePoint.infty}$ , and the homeomorphism  $\mathbb{O}^* \cong S^8$  is `onePointEquivSphereOfFinrankEq` applied to  $\dim_{\mathbb{R}} \mathbb{O} = 8$  (`Mathlib.Topology.Compactification.OnePoint.Sphere`). The compactification is what makes the value of a section at its real pole a well-defined point of  $\mathbb{O}^*$  — the point at infinity  $\infty = N$ . There is no base poset: the two worlds are built directly on  $\mathbb{O}^*$  as translation groupoids (Definition 9.6).

*Remark 9.2* (Compactifying the cited results). Most of the slice-preserving / slice-regular results cited below (GPV, AdF, Wang, Bisi–Winkelmann) are stated on uncompactified  $\mathbb{O} = \mathbb{R}^8$  or domains therein; the construction lives on  $\mathbb{O}^* = S^8$ . Each citation is used in its compactified form, obtained either directly (GPV's Riemann-manifold statements [17] are already on  $S^m$ ) or by transporting the uncompactified statement across the stereographic atlas of Definition 9.1; in the Lean development this transport is a marked derivation node over `OnePoint`, not an assumption.

*Remark 9.3* (The two north pole objects). There are two north pole objects, one in each world. In  $\mathcal{H}_1$  the north pole  $N$  of  $\mathbb{O}^*$  is a  $G_2$ -fixed object: its own connected component, with  $\text{Aut}(N) = G_2$ , carrying no universal property. In the slice world  $\mathcal{S}_2$  the north pole is the object

$$\mathbf{n} := \infty = N \text{ regarded as an object of } \mathcal{S}_2,$$

the common point at infinity of the slice Riemann spheres, fixed by band and direction alike. They are objects of *different* groupoids, related by the section through C1: the pole of  $A$  has value  $\infty$ , so the section functor carries the pole point to  $\mathfrak{n}$  (Remark 9.8). The residue field of a closed zero sorts the picture: residue- $\mathbb{R}$  (the  $G_2$ -fixed locus  $\mathbb{R} \cup \{N\}$ , classically trivial) versus residue- $\mathbb{C}$  (the 6-sphere orbits, classically nontrivial; the dictionary is Part 2 and Theorem 8.2).

## What a slice-preserving section does

**Definition 9.4** (The section's slice data and equivariance). Let  $A \in \mathcal{R}$ . By the  $I$ -independent lift (Definition 6.5; Wang [16, Rem. 2.11], Bisi–Winkelmann [4, §3.7]) there is one holomorphic stem with  $\mathbb{R}$ -valued components,

$$A(x + Iy) = F_1(x + iy) + I F_2(x + iy), \quad F_1, F_2 \text{ } \mathbb{R}\text{-valued},$$

the same for every  $I \in S^6$ . Three consequences used below:

- (i) *Slice preservation of values.*  $A(\mathbb{C}_I^*) \subseteq \mathbb{C}_I^*$  for every  $I$ : the value at a point lies on that point's own slice sphere (Definition 5.1).
- (ii)  *$G_2$ -equivariance.* For  $g \in G_2$  and non-real  $q = x + Iy$ ,

$$A(g \cdot q) = F_1 + g(I) F_2 = g \cdot (F_1 + I F_2) = g \cdot A(q),$$

since  $g \cdot (x + Iy) = x + g(I)y$  (Definition 7.3) and  $F_1, F_2$  are real; for  $q \in \mathbb{R} \cup \{N\}$  both sides are  $G_2$ -fixed, because  $A(\mathbb{R}) \subseteq \mathbb{R}$  and  $A(N) \in \bigcap_I \mathbb{C}_I^* = \mathbb{R} \cup \{N\}$  by (i). (For  $\zeta_{\mathbb{O}^*}$  this is Theorem 7.4; the computation above is the same one, run for an arbitrary intrinsic section.)

- (iii) *Blindness to the sphere direction.* On a  $G_2$ -orbit  $S_{(\sigma, \gamma)}$  the stem coordinates  $(F_1, F_2)(\sigma + i\gamma)$  are constant; in particular if  $A$  vanishes at one point of the orbit it vanishes on the whole orbit (Lemma 10.2).

No sphere-fibration and no tower enter: a slice-preserving section carries one holomorphic stem per slice, and the direction  $S^6$  is traversed only by the  $G_2$ -morphisms already present in both worlds.

*Remark 9.5* (Role of C4). Condition C4 — infinitely many residue- $\mathbb{C}$  zeros — matches the class to its infinite structure (the infinite Euler family of C2, the infinite Weierstrass divisor of C3) and gives the collapse its force: the degenerate fibre of the transport is an infinite family, so that whether it lies in one connected component of  $\mathcal{T}_A$  (Definition 9.9) is an infinite alignment, decided by Theorem 10.5. There is one connected component or there is not; nothing intermediate.

## The two worlds as translation groupoids

**Definition 9.6** (The two translation groupoids). Recall the *translation groupoid* (action groupoid) of a group action  $G \curvearrowright X$ : the category  $G \ltimes X$  whose objects are the points of  $X$  and whose morphisms  $x \rightarrow y$  are the elements  $g \in G$  with  $g \cdot x = y$ ; it is a groupoid, and  $\pi_0(G \ltimes X) = X/G$  is the orbit set (Quillen [24, §1]: components of a category  $\leftrightarrow$  components of its classifying space; in Lean, `CategoryTheory.ActionCategory`).

The *octonionic world* is the translation groupoid  $\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*$  of  $G_2 = \text{Aut}(\mathbb{O})$  acting on  $\mathbb{O}^* = S^8$  (Definitions 9.1 and 7.3; Remark 7.6). Its connected components are the  $G_2$ -orbits, of two kinds (Baez [3]; Theorem 7.5; Remark 7.6):

- the fixed points, forming the compactified real axis  $\mathbb{R} \cup \{N\}$ , of residue field  $\mathbb{R}$ ; and
- for each real  $\sigma$  and each radius  $\gamma > 0$ , the direction 6-sphere  $S_{(\sigma, \gamma)} = \{\sigma + \gamma v : v \in S^6\}$ , of residue field  $\mathbb{C}$  — the  $G_2$ -orbit of any of its points, labelled by its real centre  $\sigma$  and radius  $\gamma$ .

The *slice world*  $\mathcal{S}_2$  organizes the *codomain*  $\mathbb{O}^*$  of a section by its slice decomposition. Every slice contains the real axis by the definition of a slice,  $\mathbb{C}_I = \text{Span}_{\mathbb{R}}(1, I)$ , two slices meeting exactly in  $\mathbb{R}$  (Definition 2.4; [1, §1]), and the compactification adjoins the single shared point  $\infty = N$  (Definition 9.1; GPV [17]), so

$$\mathbb{O}^* = \bigcup_{I \in S^6} \mathbb{C}_I^*, \quad \mathbb{C}_I^* \cap \mathbb{C}_J^* = \mathbb{R} \cup \{N\} \quad (J \neq \pm I).$$

$\mathcal{S}_2$  is the groupoid whose objects are the points of  $\mathbb{O}^*$  so decomposed, and whose morphisms are generated by

- the *direction* isomorphisms:  $g : w \rightarrow g \cdot w$  for  $g \in G_2$ , carrying  $\mathbb{C}_I^*$  to  $\mathbb{C}_{g(I)}^*$  and fixing  $\mathbb{R} \cup \{N\}$  pointwise (Remark 7.6); and
- the *band* automorphism families: on each slice sphere  $\mathbb{C}_I^*$ , the phases  $u_\theta : z \mapsto e^{I\theta} z$  — the Möbius self-maps of  $S_I^2$  fixing the value-origin 0, the point at infinity  $N$ , and the modulus.

The objects 0 and  $N$  are fixed by every generating morphism. No identification of distinct objects is made here: the two worlds share the underlying points of  $\mathbb{O}^*$  and differ in their acting structure ( $G_2$  alone, versus direction together with the band). This is exactly the structure the base  $\mathcal{B}$  already carries (Definition 9.9); nothing is added there.

## The section functor

**Theorem 9.7** (The section functor  $\Phi$ ). *For a section  $A \in \mathcal{R}$ , the assignment*

$$\Phi(q) = A(q) \text{ on objects,} \quad \Phi(g : q \rightarrow g \cdot q) = (g : A(q) \rightarrow A(g \cdot q)) \text{ on morphisms,}$$

*is a functor of groupoids  $\Phi : \mathcal{H}_1 \rightarrow \mathcal{S}_2$  — the section functor.*

*Proof.* This is the standard fact that an equivariant map of  $G$ -sets induces a functor of the associated translation groupoids, applied to  $A$ . On objects,  $A(q)$  is a point of  $\mathbb{O}^*$  lying on the slice sphere of  $q$  (Definition 9.4(i)), hence an object of  $\mathcal{S}_2$ . On morphisms,  $G_2$ -equivariance  $A(g \cdot q) = g \cdot A(q)$  (Definition 9.4(ii)) says precisely that  $g$  is a direction morphism  $A(q) \rightarrow A(g \cdot q)$  of  $\mathcal{S}_2$  (Definition 9.6), so  $\Phi(g)$  is well-typed. Functoriality is immediate:  $\Phi(e) = \text{id}_{A(q)}$ , and for composable  $g, h$  the composite of the direction morphisms  $g : A(q) \rightarrow A(gq)$  and  $h : A(gq) \rightarrow A(hgq)$  is  $hg : A(q) \rightarrow A(hgq) = \Phi(hg)$ . On each slice the underlying value assignment is the holomorphic self-map  $A_v$  of  $S_v^2$ , compatible with the ring structure because the regular product restricts to ordinary multiplication on the commutative slice (Theorem 5.2, Proposition 5.3).  $\square$

*Remark 9.8* (What  $\Phi$  collapses, and the cone from the pole). The collapse belongs to this functor, not to the codomain. By Definition 9.4(iii),  $\Phi$  carries each residue- $\mathbb{C}$  zero-sphere of  $A$  — a 6-sphere component of  $\mathcal{H}_1$  (Lemma 10.2) — onto the *single* object  $0 \in \mathcal{S}_2$ , the value-origin, fixed by band and direction; and by C1 it carries the pole point onto the single object  $\mathfrak{n}$  (Remark 9.3). Thus the pole pulls a cone in the value world: every path of  $\mathcal{H}_1$  into the pole maps to a value path into the slice-world north pole. The differing centres and radii of the zero-spheres are not remembered by  $\Phi$ 's object map; they are remembered by the *transport* over the base (Definition 9.9), which is where the theorem reads them.

## The base of the degenerate set and the Grothendieck construction

**Definition 9.9** (The base  $\mathcal{B}$  of the degenerate set, the band  $F$ , and  $\mathcal{T}_A = \int_{\mathcal{B}} F$ ). *The base.* The exponential's degenerate fibre over a real value is the level-indexed family of Lemma 6.3:  $\exp^{-1}(-r) = \{\log r + I(2k+1)\pi\}$ , one real level  $\log r$ , winding as band data — a structure of the space itself, absent over  $\mathbb{C}$  ([17, Rem. 5.2(b)]). The *base*  $\mathcal{B}$  is the groupoid of these levels: one object for each real level  $c$ , with the direction automorphisms  $G_2$  at each level. GPV's  $\log |\cdot|$  is a label on objects, never an operation. Distinct levels are distinct objects with *no* morphisms between them, so the level is constant along every zigzag and  $\pi_0(\mathcal{B})$  is the set of levels by construction — connectivity here is the groupoid's (reachability by declared morphisms, Definition 9.6), not the ambient topology of  $\mathbb{O}$ .

*The band.* Let  $F : \mathcal{B} \rightarrow \text{Grpd}$  send each level to the band groupoid  $U(1)$  (one object; automorphism group the phases) and each direction morphism to the identity of the band. The winding index  $2k+1$  of the fibre is carried by the band as an automorphism datum within a component — the loops of  $F(c)$  realize the winding number  $\omega = |\sigma^c|/2$  of the lifted value-loops ([21, Cor. 5.21]; Proposition 6.7) — and is never an object label of  $\mathcal{B}$ .

*The total object.* The *Grothendieck construction* of  $F$  is the category

$$\mathcal{T}_A = \int_{\mathcal{B}} F,$$

with objects the pairs  $(c, x)$ ,  $x$  in the band over  $c$ , and morphisms the pairs  $(f, \varphi)$  moving the base and transporting the fibre (Thomason [23]; Vistoli [26, §3.1]; Stacks [18, Tag 0049]; in Lean, `CategoryTheory.Grothendieck`). By Thomason's theorem its classifying space models the homotopy colimit,  $B(\int_{\mathcal{B}} F) \simeq \text{hocolim}_{\mathcal{B}} (B \circ F)$  [23, Thm. 1.2] (statement as quoted in [22, Thm. 1.1]; textbook treatment Riehl [25, Ch. 5–6]); the proof below uses only its  $\pi_0$  shadow (Lemma 10.3).

*How the section populates the diagram.* The transport data over  $\mathcal{B}$  is supplied by the winding lift of the section's value-paths: a lift of a value-loop  $\gamma$  of  $A$  is a  $\Gamma$  with  $\text{pr}_1 \circ \Gamma = \gamma \circ \exp$  (Theorem 6.6; [21, Def. 5.11, Prop. 4.2]); it is *unique* when the loop is tame — unique companion — ([21, Def. 4.7]) and *exists, and is itself a loop*, exactly when the signature criterion  $\sigma \in \{0, -1\}$  holds on each obstruction interval ([21, Cor. 5.13]; the earlier citation to Cor. 5.22 is superseded by this pinned number). The residue- $\mathbb{C}$  zero-spheres of  $A$  appear in  $\mathcal{T}_A$  as the *degenerate fibre* of this transport — an output of the construction, never an input — and the theorem computes the image of that fibre in  $\pi_0(\mathcal{T}_A)$ .

## 10 The concentricity theorem

**Proposition 10.1** (Slice-regular Weierstrass factorization; the content of C3). *Let  $A \in \mathcal{R}$  have a zero of order  $m \geq 0$  at the origin and no non-real isolated zeros (for  $A \in \mathcal{R}$  the latter is automatic: non-real zeros come in whole spheres, Lemma 10.2). Then on  $\mathbb{O}^* \setminus \{N\}$  it factors as*

$$A = q^m R S h,$$

where  $q$  denotes the octonionic coordinate function (the identity section, so  $q^m$  carries the order- $m$  zero at the origin),  $R$  is a slice-preserving factor vanishing exactly at the real zeros of  $A$ ,  $S$  is a factor vanishing exactly at its spherical (residue- $\mathbb{C}$ ) zero-spheres, and  $h$  is never vanishing; each of  $R, S$  is the slice-regular Weierstrass product over the corresponding zeros. The factorization holds including the case where either family of zeros is infinite.

*Proof.* This is the factorization theorem of Altavilla–de Fabritiis [2, Prop. 3.1, Thm. 3.2], whose primary factors are the slice-regular ones of Gentili–Vignozzi [10]; convergence of the infinite products is part of the cited statement. The pair  $R, S$  is the residue split named in C3:  $R$  carries the residue- $\mathbb{R}$  (classically trivial) zeros and  $S$  the residue- $\mathbb{C}$  (classically nontrivial) ones, the latter infinite in number by C4.  $\square$

**Lemma 10.2** (Residue- $\mathbb{C}$  zeros are 6-sphere components of  $\mathcal{H}_1$ ). *For  $A \in \mathcal{R}$ , the non-real zeros of  $A$  are 6-spheres, each a single connected component of the translation groupoid  $\mathcal{H}_1$ .*

*Proof.*  $A$  is slice-preserving, so on a sphere  $\sigma + \gamma S^6$  ( $\gamma > 0$ ) it has the form  $A(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v\beta(\sigma, \gamma)$  with  $\alpha, \beta$  real (the real stem, [16]); since  $v$  is an imaginary unit,  $\alpha + v\beta = 0$  forces  $\alpha = \beta = 0$ , independently of  $v$ . Hence a non-real zero occurs on a *whole* sphere  $\sigma + \gamma S^6$  or at none of its points (the spherical-zero structure, [1]); real zeros are isolated real points (residue field  $\mathbb{R}$ ). Each such sphere  $\sigma + \gamma S^6$  is the  $G_2$ -orbit of any of its points (Theorem 7.5:  $G_2$  acts transitively on  $S^6$  with stabiliser  $SU(3)$ ), and the connected components of  $\mathcal{H}_1$  are exactly the  $G_2$ -orbits — in a groupoid two objects share a component iff a morphism joins them, and a morphism  $x \rightarrow y$  of  $\mathcal{H}_1$  is a  $g \in G_2$  with  $g \cdot x = y$ . Therefore the residue- $\mathbb{C}$  zeros — the non-real ones — are exactly the 6-sphere components of  $A^{-1}(0)$ .  $\square$

**Lemma 10.3** ( $\pi_0$  of a Grothendieck construction). *For a functor  $F : \mathcal{B} \rightarrow \mathbf{Grpd}$ , the connected-components functor carries the Grothendieck construction to the colimit of the component diagram:*

$$\pi_0\left(\int_{\mathcal{B}} F\right) \cong \operatorname{colim}_{\mathcal{B}} (\pi_0 \circ F).$$

*Proof.* The functor  $\pi_0 : \mathbf{Cat} \rightarrow \mathbf{Set}$  is left adjoint to the inclusion of discrete categories, so it preserves colimits; and components of a category correspond one-to-one with components of its classifying space (Quillen [24, §1]). The identity is also direct at the level of categories. An object of  $\int_{\mathcal{B}} F$  is a pair  $(b, x)$ ; a morphism  $(f, \varphi)$  projects to a morphism  $f$  of  $\mathcal{B}$  and restricts, over a fixed base object, to a morphism of the fibre. Hence a zigzag joining  $(b, x)$  to  $(b', x')$  projects to a zigzag joining  $b$  to  $b'$  and, fibrewise, joins the band coordinates; two objects of  $\int_{\mathcal{B}} F$  lie in one component exactly when their base objects lie in one component of  $\mathcal{B}$  and their fibre coordinates are identified there, which is precisely  $\operatorname{colim}_{\mathcal{B}} (\pi_0 \circ F)$ . Passing to classifying spaces, this is the  $\pi_0$  of Thomason's equivalence  $B(\int_{\mathcal{B}} F) \simeq \operatorname{hocolim}_{\mathcal{B}} (B \circ F)$  [23] (see also [25, Ch. 5–6]).  $\square$

**Definition 10.4** ( $A$ -sections). An  $A$ -section is a section  $A$  of the ring  $\mathcal{R}$  of slice-preserving slice-regular functions on  $\mathbb{O}^* = S^8$  (Definition 5.1) — equivalently, a slice-preserving *semiregular* function on  $\mathbb{O}^*$  (the meromorphic slice class; [1, §11]) — with the following four properties.

- (C1) *Meromorphic continuation; single simple pole.*  $A$  is meromorphically continued to  $\mathbb{O}^*$  with exactly one pole; it is simple, lies at a real point, and its value there is the point at infinity  $\infty = N$ .
- (C2) *Infinite Euler product.* On a slice right half-space  $\Omega_0 \subset \mathbb{O}^*$ ,  $A = \exp(\sum_p \ell_p)$  for an *infinite* summable family  $\{\ell_p\}$  of slice-preserving slice-regular functions each zero-free on  $\Omega_0$ ; in particular  $A$  is zero-free on  $\Omega_0$ .
- (C3) *Infinite Weierstrass factorization.* On  $\mathbb{O}^* \setminus \{\text{pole}\}$ ,

$$A = q^m R e^g \prod_n \mathcal{E}(\cdot; q_n),$$

where  $q$  is the octonionic coordinate,  $m \geq 0$ ,  $R$  is the slice-regular Weierstrass product over the residue- $\mathbb{R}$  (real) zeros of  $A$ , vanishing only on  $\mathbb{R}$ ,  $g$  is slice-preserving entire,  $\{q_n\}$  enumerates the residue- $\mathbb{C}$  zero-spheres of  $A$ , and  $\mathcal{E}(\cdot; q_n)$  is the slice-regular Weierstrass primary factor of  $q_n$  (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]); the factorization holds with either family infinite.

(C4) *Infinitely many residue- $\mathbb{C}$  zeros.* The index set  $\{q_n\}$  is infinite. A closed zero of such a section is a real point (residue field  $\mathbb{R}$ ) or a 6-sphere  $S_{(\sigma, \gamma)}$  (residue field  $\mathbb{C}$ ; [2]); residue- $\mathbb{C}$  names the classically nontrivial zeros (Part 2).

**Theorem 10.5** (Concentricity). *Let  $A$  be an  $A$ -section (Definition 10.4). Then the residue- $\mathbb{C}$  zero spheres of  $A$  all lie in a single connected component of the total object  $\mathcal{T}_A$  (Definition 9.9) — equivalently, they have one and the same image in  $\pi_0(\mathcal{T}_A)$ .*

*Proof.* Let  $p : \mathcal{T}_A = \int_{\mathcal{B}} F \rightarrow \mathcal{B}$  be the Grothendieck construction of Definition 9.9. Hypotheses C1–C4 assemble the diagram; the residue- $\mathbb{C}$  zero-spheres then arrive as the degenerate fibre of its transport, an output; and  $\pi_0$  reads off their component.<sup>1</sup>

*Assembly: the unique tame transport extending the exponential's base.* The exponential's degenerate fibre over a real value is the level-indexed family of Lemma 6.3 — one real level, winding carried by the band — a structure of the space itself, absent over  $\mathbb{C}$  ([17, Rem. 5.2(b)]; Definition 9.9). C2 and C3 extend the exponential structure over  $A$ : by C2, on  $\Omega_0$  the section is the exponential  $A = \exp(\sum_p \ell_p)$  of an infinite summable slice-preserving family, zero-free there, so the continuation of the hypercomplex logarithm of  $A$  exists outright on  $\Omega_0$  (Theorem 6.6: a continuation is  $\tilde{\gamma}$  with  $\exp \circ \tilde{\gamma} = \gamma$ ); by C3, on  $\mathbb{O}^* \setminus \{\text{pole}\}$  the section is the exponential expression  $A = q^m Re^g \prod_n \mathcal{E}(\cdot; q_n)$  over its full divisor. On any slice  $\mathbb{C}_I$  both are the one holomorphic stem  $A_I$  (Definition 9.4; Wang [16, Rem. 2.11]), agreeing on the overlap  $\Omega_0$ , hence coinciding wherever both are defined (Identity Theorem 3.3). The value-loops of  $A$  are therefore tame — unique companion ([21, Def. 4.7]) — so their lifts to the logarithm manifold, when they exist, are unique; and by C1 the section has exactly one pole, whose value is  $\mathbf{n}$  (Remark 9.3), the cone through which the value-loops close: along these loops the signature criterion holds and the lift exists *and is itself a loop* ([21, Cor. 5.13]). This is the transport of Definition 9.9, canonically attached to  $A$  by C1–C4 jointly. Its degenerate fibre — the exp-preimage data over the real values met by the unique tame lift — is the level-indexed fibre of Lemma 6.3, and the residue- $\mathbb{C}$  zero-spheres  $\{q_n\}$  of C3 are exactly this degenerate fibre, infinite by C4.

Through the commuting triangle  $\pi \circ E = \exp$  ([17, Rem. 5.2(a)]), the unique tame lift traverses the logarithm manifold as a single closed loop ([21, Cor. 5.13]), and every point of the degenerate fibre it meets is, by Lemma 6.3, the level  $\log r$  paired with an odd winding height  $I(2k+1)\pi$ : all multiplicity in the fibre lies in the winding direction ([21, Cor. 5.21]), none in the level. Since  $\mathcal{B}$  is static — no morphisms between distinct levels (Definition 9.9) — the level is a conserved quantity along every zigzag of  $\mathcal{T}_A$ , and the degenerate fibre of the unique tame transport attached to the  $A$ -section — the residue- $\mathbb{C}$  zero-spheres  $\{q_n\}$  of C3 — lies over a *single* level.

*The cocartesian computation of  $\pi_0$ .* The fibre  $F$  is connected, so by Lemma 10.3

$$\pi_0(\mathcal{T}_A) \cong \operatorname{colim}_{\mathcal{B}}(\pi_0 \circ F) \cong \pi_0(\mathcal{B}),$$

under which a residue- $\mathbb{C}$  zero-sphere maps to the class of the base object beneath it, and two of them share a component of  $\mathcal{T}_A$  if and only if they share that class. By the assembly and the placement above, the degenerate fibre lies over one and the same class — the zero-spheres over a

<sup>1</sup>Remark 10.6 sketches an alternative, finality-based computation of  $\pi_0$ ; it is expository and not used here.

single real level  $c$ , one for each winding index carried by the band — hence has a single image in  $\pi_0(\mathcal{T}_A)$ : one connected component.

*Readback in  $\mathcal{H}_1$ .* By Lemma 10.2 each residue- $\mathbb{C}$  zero-sphere is a  $G_2$ -orbit component of  $\mathcal{H}_1$ , carried into the slice world by the section functor  $\Phi$  (Theorem 9.7; Remark 9.8); distinct spheres are distinct orbits, hence pairwise disjoint, and by C4 there are infinitely many. The theorem states only the one-component conclusion; reading the component as a common *centre* is the downstream translation of Corollary 10.9.  $\square$

*Remark 10.6* (A second proof, via finality). The single-component conclusion also follows from Quillen’s §1 [24], on the same assembly, in either of two forms. *Finality:* by C1 the continuation of  $A$  carries every zero-bearing level of  $\mathcal{B}$  to the north pole  $\mathfrak{n}$ , and the winding lift supplies a unique tame companion along each such continuation ([21, Def. 4.20, Cor. 5.13]); so the comma category from a zero-bearing level to  $\mathfrak{n}$  is non-empty and connected, and  $\mathfrak{n}$  is final over the zero-bearing part of  $\mathcal{B}$  — the  $\pi_0$  shadow of Theorem A (“if  $Y \setminus f$  is contractible for every  $Y$ , then  $f$  is a homotopy equivalence”; connected and non-empty is the  $\pi_0$ -level hypothesis; textbook form Riehl [25, §8.3, §8.5]). *Fibred form:* the projection  $p : \int_{\mathcal{B}} F \rightarrow \mathcal{B}$  is precofibred with fibre the connected band, and Quillen’s corollary to Theorem A (pre(co)fibred with contractible fibres  $\Rightarrow$  homotopy equivalence) has as its  $\pi_0$  shadow exactly the bijection  $\pi_0(\mathcal{T}_A) \cong \pi_0(\mathcal{B})$  used above. Thomason’s theorem [23] lifts the identity of Lemma 10.3 to the classifying-space equivalence  $B(\int_{\mathcal{B}} F) \simeq \text{hocolim}_{\mathcal{B}}(B \circ F)$ . The cocartesian argument above is the primary one; this is recorded as an alternative.

**Theorem 10.7** (Connected is concentric: the dictionary). *Residue- $\mathbb{C}$  spheres of the degenerate family of Lemma 6.3 lie in one connected component of the static base — have one image in  $\pi_0(\mathcal{B})$  — if and only if they share one real level, if and only if they are 6-spheres about one real centre: concentric.*

*Proof.* The components of the static base are the real levels by construction: the base is static, no morphisms between distinct levels, the level conserved along every zigzag (Definition 9.9); so one image in  $\pi_0(\mathcal{B})$  is one level. By Lemma 6.3 the degenerate fibre over  $-r$  is  $\exp^{-1}(-r) = \{\log r + I(2k+1)\pi\}$  — one real level  $\log r$ , winding carried by the band — and its spheres have centre  $\log r$  and radii  $(2k+1)\pi$ : one level is one centre. Spheres about one centre are concentric. No property of  $A$ -sections and no part of Theorem 10.5 enters.  $\square$

*Remark 10.8* (Dictionary vs. application). The dictionary above is logically independent of the collapse: its **uses** carries no arrow from Theorem 10.5, and the blueprint graph shows the two nodes unlinked. The application, Corollary 10.9, consumes both — the collapse supplies *one component*; the dictionary reads it as *one centre*.

**Corollary 10.9** (Translation to the classical framework). *Identifying residue- $\mathbb{C}$  zeros with the classically nontrivial zeros, and residue- $\mathbb{R}$  zeros with the trivial ones (Part 2), such a section has its nontrivial zeros realised as infinitely many pairwise disjoint concentric 6-spheres about a single real centre.*

*Proof.* By Theorem 10.5 the residue- $\mathbb{C}$  zero-spheres lie in one connected component of  $\mathcal{T}_A$ . By the dictionary (Theorem 10.7) one component of the static base is one real level, hence one real centre  $c$ : the spheres are concentric about  $c$ . Their identification with the classical nontrivial zeros is Theorems 8.1 and 8.2; infinitude and pairwise disjointness are C4 and Lemma 10.2.  $\square$

## 11 Corollaries

**Corollary 11.1** ( $\zeta_{\mathbb{O}^*}$  is an  $A$ -section).  $\zeta_{\mathbb{O}^*}$  is an  $A$ -section (Definition 10.4): a section of  $\mathcal{R}$  satisfying (C1)–(C4).

*Proof.*  $\zeta_{\mathbb{O}^*} \in \mathcal{R}$  by slice-preservation (Theorem 7.2). The verification of (C1)–(C4) cites the classical facts of the Riemann zeta function (Part 1) and the slice-regular zero-geometry that translates them:

- **(C1)** *Meromorphic continuation; simple pole.* Classical fact:  $\zeta$  is meromorphically continued with a single simple pole at  $s = 1$  (Theorem 1.1); the value of  $\zeta_{\mathbb{O}^*}$  there is the point at infinity,  $\zeta_{\mathbb{O}^*}(1) = \infty = N$  (Definition 4.1).
- **(C2)** *Infinite Euler product.* Classical fact:  $\zeta = \prod_p (1 - p^{-s})^{-1} = \exp(\sum_p \ell_p)$  on  $\Re s > 1$ , the family indexed by the infinitely many primes (Theorem 1.2).
- **(C3)** *Infinite Weierstrass factorization.* Supplied by the *slice-regular* Weierstrass factorization (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]):  $\zeta_{\mathbb{O}^*} = q^m \text{Re}^g \prod_n \mathcal{E}(\cdot; q_n)$ , where  $m = 0$  ( $\zeta(0) = -\frac{1}{2} \neq 0$ ),  $R$  is the product over the residue- $\mathbb{R}$  zeros — the classically trivial zeros  $-2, -4, \dots$ , honest real zeros of  $\zeta_{\mathbb{O}^*}$  — and  $\{q_n\}$  enumerates its residue- $\mathbb{C}$  zero-spheres.
- **(C4)** *Infinitely many residue- $\mathbb{C}$  zeros.* Classical fact:  $\zeta$  has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). By the residue dictionary below these are exactly the residue- $\mathbb{C}$  zeros, so  $\{q_n\}$  is infinite.

The one remaining classical fact enters the downstream centre-pinning rather than (C1)–(C4): the *functional equation*  $\xi(s) = \xi(1-s)$ ,  $\xi(\bar{s}) = \overline{\xi(s)}$  (Theorem 1.1), used in Corollary 11.2. The *residue dictionary* — a closed zero of  $\zeta_{\mathbb{O}^*}$  is a real point (residue field  $\mathbb{R}$ , the classically *trivial* zeros) or a 6-sphere  $S_{(\sigma, \gamma)}$  (residue field  $\mathbb{C}$ , the classically *nontrivial* zeros) — is the slice-regular zero-geometry of Theorem 8.2 and [1]; it is what makes “residue- $\mathbb{C}$ ” and “nontrivial” the same notion. Thus (C1)–(C4) hold.  $\square$

**Corollary 11.2** (Riemann Hypothesis). *Every nontrivial zero of the classical Riemann zeta function has real part  $\frac{1}{2}$ ; in particular infinitely many zeros lie on the line  $\Re s = \frac{1}{2}$ .*

*Proof.* By Corollary 11.1,  $\zeta_{\mathbb{O}^*}$  satisfies (C1)–(C4); by Theorem 10.5 the residue- $\mathbb{C}$  zero-spheres of  $\zeta_{\mathbb{O}^*}$  lie in a single connected component of  $\mathcal{T}_{\zeta_{\mathbb{O}^*}}$ . By Corollary 10.9 that one component is a single real centre  $c$ , so the residue- $\mathbb{C}$  zeros are infinitely many concentric 6-spheres about  $c$ ; under the residue dictionary (Corollary 11.1) these are the classically nontrivial zeros, so every nontrivial zero has real part  $c$ . The functional equation  $\xi(s) = \xi(1-s)$  (Theorem 1.1) sends a zero of real part  $c$  to one of real part  $1-c$ ; applied to the *common* centre it forces  $c = 1-c$ , whence  $c = \frac{1}{2}$  (this is Theorem 8.3 read on  $\zeta_{\mathbb{O}^*}$ ). Therefore every nontrivial zero has real part  $\frac{1}{2}$ , and by (C4) infinitely many lie on  $\Re s = \frac{1}{2}$ .  $\square$

*Remark 11.3* (Status and formalization). Theorem 10.5 rests on the cited slice-regular inputs — the Weierstrass factorization (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]), the winding lift and its criteria (Theorem 6.6, Proposition 6.7; [21, Def. 5.11, Def. 4.20, Cor. 5.13, Cor. 5.21]), and the exponential’s degenerate fibre (Theorem 6.1; Lemma 6.3) — together with the identity  $\pi_0(\mathcal{T}_A) \cong \text{colim}_{\mathcal{B}}(\pi_0 \circ F)$  of Lemma 10.3. A second proof from Quillen’s §1 (Theorem A and its pre(co)fibred corollary [24]; Riehl [25, §8.3]) is recorded in Remark 10.6. The placement sentence remains the author’s and is conspicuously marked inside the proof of Theorem 10.5; the static-base reading is fixed, so the sentence must speak of levels and winding only.

The translation from “one connected component” to “one real centre,” and thence to concentric 6-spheres and the classical nontrivial zeros, is Corollary 10.9: the components of the static base are the real levels (Definition 9.9), and the equivalence Theorems 8.1 and 8.2 perform the identification. It is a corollary of the theorem, kept logically separate: the Concentricity Theorem itself states and uses only the one-component conclusion.

A machine formalization is future work; nothing in this document has been run in Lean. The categorical objects rest on existing Mathlib foundations — the translation groupoid  $\mathcal{H}_1$  (`CategoryTheory.ActionCate`), the Grothendieck construction  $\mathcal{T}_A$  (`CategoryTheory.Grothendieck`), and  $\pi_0$  (`CategoryTheory.ConnectedCompon`) — while the octonionic and  $G_2$  data and the band  $U(1)$ , absent from Mathlib, enter as axioms carrying their citations [19]. The section functor  $\Phi$  (Theorem 9.7) and the winding lift (Theorem 6.6) are the constructions to formalize; the one-component conclusion is then the  $\pi_0$  computation of Lemma 10.3.

## References

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